

MUCOSOLVAN®

24 Hours

Composition

1 sustained-release capsule contains 75 mg trans-4-[(2-amino-3,5-dibromo-benzyl) amino] cyclohexanol hydrochloride (= ambroxol hydrochloride)
Excipients:
Crospovidone, carnauba wax, stearyl alcohol, magnesium stearate

Capsule casing:
Gelatine, Purified water, Titanium dioxide (E 171), Iron oxide (E 172)

Capsule marking:
Shellac, Titanium dioxide (E 171)

Description

Oblong hard gelatin capsules consisting of red opaque cap and an orange opaque body; the cap is printed with “MUC 01” in white. Capsule contents : round, yellowish white pellets with a smooth, shiny surface, mixed with a small quantity of powder.
Dimensions:
Capsule size : 2
Diameter of capsule cap : 6.0-6.4mm
Length of capsule : 17.4 – 18.4mm

Pharmacological Properties Pharmacodynamics

Pharmacotherapeutic group:
Mucolytics
ATC code: R05CB06
In preclinical investigations, ambroxol hydrochloride, the active ingredient of MUCOSOLVAN, has been shown to increase the proportion of serous bronchial secretion.

Ambroxol hydrochloride also increases the production of surfactant by acting directly on type II pneumocytes in the alveoli and Clara cells in bronchioles and simulating the activity of the cilia of the ciliated epithelium. These effects result in reduced viscosity of the mucus and improved transport (mucociliary clearance). Improved mucociliary clearance has been proven in clinical pharmacological studies. The increased production and reduced viscosity of secretions and improved mucociliary clearance support expectoration and facilitate the coughing up of phlegm.

Long-term use(6 months) of Mucosolvan® (Mucosolvan® sustained-release capsules 75 mg) in COPD patients led to substantially reduced exacerbations after a treatment period of 2 months. Patients treated with Mucosolvan® missed significantly fewer days of work due to illness and the duration of treatment with antibiotics was reduced. In comparison with a placebo, treatment with Mucosolvan® sustained-release capsules showed a statistically significant improvement in symptoms with regard to expectoration problems, coughing, dyspnoea, auscultatory signs. The local anaesthetic effect of ambroxol hydrochloride was studied on the eyes of rabbits. It is likely due to its blocking effect on the sodium channels. Ambroxol hydrochloride blocks the hyperpolarized channels in cloned voltage-dependent neuronal sodium channels *in vitro*. The bond was reversible and concentration-dependent. It was determined *in vitro* that ambroxol hydrochloride has an anti-inflammatory effect. Thus, ambroxol hydrochloride significantly reduced the cytokine release of mononuclear and polymorphonuclear cells in the blood and tissue. Clinical studies in patients with a sore throat demonstrated that ambroxol hydrochloride administered as a 20 mg lozenge significantly reduced pain and redness in the throat. These pharmacological properties are consistent with the additional observation of faster pain relief that was achieved with inhalative ambroxol treatment for upper respiratory conditions in clinical efficacy studies. Concentrations of the antibiotics amoxicillin, cefuroxime, erythromycin and doxycycline in the sputum and bronchial secretions increase after the use of ambroxol hydrochloride. To date, this has not been shown to have any clinical relevance.

Antiviral properties in *in vitro* and in animal models:
In *in vitro* studies in human tracheal epithelial cells a reduction of rhinovirus (RV 14) replication has been observed. In a mouse airway model, a reduction of Influenza A virus replication was observed with ambroxol pretreatment. To date, this has not been shown to have any clinical relevance.

Pharmacokinetics

Absorption:
Absorption of immediate release oral dosage forms of ambroxol hydrochloride is fast and complete, with dose linearity in the therapeutic range. Maximum plasma levels are reached within 1 to 2.5 hours after administration of the immediate – release formulation and after a median of 6.5 hours after administration of the slow-release formulation.

The absolute bioavailablility after ingesting a 30 mg tablet is 79%. The sustained-release capsule showed a relative availability of 95% (dose-normalized) in comparison to tablets with unchanged active substance release (60 mg daily dose, 30 mg twice daily).

Food was not found to have any effect on the bioavailability of ambroxol hydrochloride.

Distribution:
Distribution of ambroxol hydrochloride from blood to tissue is rapid and pronounced, with the highest concentration of the active substance found in the lungs. The estimated volume of distribution after oral administration was estimated to be 552 litres.

In the therapeutic range, the plasma protein binding is approx. 90%.

Metabolism:
About 30% of an orally administered dose is eliminated via first pass metabolism. Ambroxol hydrochloride is primarily metabolized in the liver through glucuronidation and cleavage to dibromoanthranilic acid (approx. 10% of the dose). Studies in human liver microsomes have shown that CYP3A4 is responsible for the metabolism of ambroxol hydrochloride to dibromoanthranilic acid.

Elimination:
After 3 days of oral administration, ambroxol hydrochloride is eliminated renally, approx. 6% unchanged and approx. 26% in the form of its conjugates. The terminal elimination half-life of ambroxol hydrochloride is about 10 hours. Total clearance is in the range of 660 ml/min, with renal clearance accounting for approx. 8% of the total clearance. After 5 days an estimated 83% of the total dose (radioactively marked) is eliminated with the urine.

Special patient groups:
The elimination of ambroxol hydrochloride is reduced in patients with impaired liver function, resulting in plasma levels that are approx. 1.3 to 2 times higher. Due to the broad therapeutic range of the active substance, a dose adjustment is not necessary. Age and gender were not found to influence the pharmacokinetics of ambroxol hydrochloride to a clinically relevant degree. Therefore, deviating from the recommended dose is not necessary.

Toxicology
Single dose toxicity
Ambroxol has a low index for acute toxicity.

Repeat dose toxicity
Oral use: No toxicological target organs were discovered in repeated-dose studies in rats (52 and 78 weeks), rabbits (26 weeks), mice (4 weeks) or dogs (52 weeks). The “no observed adverse effect level” (NOAEL) was 50 mg/kg/day in rats, 40 mg/kg/day in rabbits, 150 mg/kg/day in mice and 10 mg/kg/day in dogs. Intravenous use: Toxicity studies with ambroxol hydrochloride showed no severe local or systemic toxicity including histopathology over 4 weeks in rats (4, 16 and 64 mg/kg/day [infusions 3 hours/day]) or in dogs (45, 90 and 120 mg/kg/day [infusions 3 hours/day]). All adverse events were reversible.

Reproductive and developmental toxicity
Ambroxol hydrochloride was neither embryotoxic nor teratogenic at tested oral doses of up to 3,000 mg/kg/day in rats and up to 200 mg/kg/day in rabbits. Fertility in male and female rats was not impaired at doses of up to 1,500 mg/kg/day. The NOAEL was 50 mg/kg/day in a study on perinatal and postnatal development. Ambroxol hydrochloride was slightly toxic for mother animals and young animals at 500 mg/kg/day (delayed development of body weight and reduced litter size).

Genotoxicity
In vitro (Ames and chromosome aberration tests) and in vivo (micronucleus test in mice) studies on genotoxicity revealed no genotoxic potential for ambroxol hydrochloride. **Carcinogenicity**
Ambroxol hydrochloride showed no tumorigenic potential in studies on carcinogenicity in mice (50, 200 and 800 mg/kg/day) or rats (65, 250 and 1,000 mg/kg/day) when added to food over a period of 105 and 116 weeks respectively.

Indications
Secretolytic therapy in acute and chronic bronchopulmonary diseases associated with abnormal mucus secretion and impaired mucus transport.

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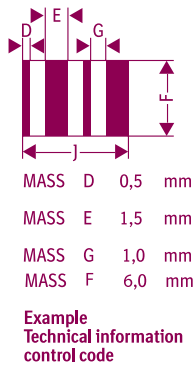
Digital · Packaging · Development

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File information	
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Plant PM code:	10054540
Second Plant PM code:	324898-07
Version of artwork:	V3
PM type:	PI
Market:	SG
Format:	160 x 580 mm
Issue date of artwork:	07/Oct/2025
Print colors:	Black
Number of print colors: 1	
Used font:	Gotham
Min. font size:	9 pt

Technical colors		
Diecut-Legendcase	Free area	Glue points

ADDITIONAL REQUIREMENT OF PACKAGING LINE	
Description :	PI MUCOSOLVAN 24H 75MG CAPCR BL5X10 / BL10 M30 SG
Dimension :	160 x 580 mm
No. of code :	491
Ref. drawing :	PR34
Issue date ofTD:	05/06/2025



Dosage and Administration

Adult : 1 sustained-release capsules once daily.
The capsules should not be opened or chewed, but swallowed whole with ample liquid. The “carrier pellets” which are occasionally present in the stools have released the active substance during their passage through the digestive system and are therefore without significance. In acute respiratory indications, medical advice should be sought if symptoms do not improve or worsen in the course of therapy. Mucosolvan can be taken with or without food.

Contraindications

Hypersensitivity to the active substance (ambroxol hydrochloride) or any of the other excipients listed. MUCOSOLVAN once a day sustained-release capsules are not suitable for children under 12 years of age on account of their strength. MUCOSOLVAN paediatric syrup 30 mg/5 ml and MUCOSOLVAN drops 30 mg/2 ml are available for use in this age group.

Special warnings and precautions

There have been reports of severe skin reactions such as erythema multiforme, Stevens-Johnson syndrome (SJS)/ toxic epidermal necrolysis (TEN) and acute generalised exanthematous pustulosis (AGEP) associated with the administration of ambroxol hydrochloride. In most cases, these could be explained by the severity of the underlying disease or concomitant administration of another drug. In the early stages of such severe skin reactions, initially only nonspecific flu-like symptoms appear, e.g. fever, arthralgia, runny nose, cough, and sore throat. If symptoms or signs of a progressive skin rash (sometimes associated with blisters or mucosal lesions) are present, ambroxol hydrochloride treatment should be discontinued immediately and medical advice should be sought. In patients with impaired bronchial motility and copious secretions (as seen, for instance, in the rare and malignant primary ciliary dyskinesia), MUCOSOLVAN once a day sustained-release capsules must be used with caution because of the risk that it may promote accumulation of secretions.

In patients with impaired kidney function or severe liver disease, MUCOSOLVAN should not be used except on medical advice. With ambroxol, as with any medicine that is metabolized hepatically and excreted renally, accumulation of the metabolites of ambroxol which are formed in the liver can be expected to occur in patients with severe renal failure.

Interactions

If Mucosolvan is used in combination with antitussives in patients with pre-existing respiratory diseases such as cystic fibrosis or bronchiectasis who are affected by mucus hypersecretion, a reduced cough reflex may lead to (serious) accumulation of mucus.

Fertility, pregnancy and Lactation Pregnancy

Ambroxol hydrochloride crosses the placental barrier. Nonclinical studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/ foetal development, labour and delivery or postnatal development. Extensive clinical experience after the 28th week of pregnancy has shown no evidence of harmful effects on the foetus. Nonetheless, the usual precautions regarding the use of drugs during pregnancy should be observed. The use of MUCOSOLVAN is not recommended during the first trimester of pregnancy in particular.

Breast-feeding

Ambroxol hydrochloride has been shown to pass into breast milk in animal studies. Use while breast-feeding is not recommended.

Fertility

Non-clinical studies do not indicate direct or indirect harmful effects with respect to fertility.

Effects on ability to drive and use machines

There is no evidence from postmarketing data for an effect on the ability to drive or use machines. Studies on the effects on the ability to drive and use machines have not been performed.

Side Effects

The following categories are normally used when reporting the frequencies of side effects:

Very common:	≥ 1/10
Common:	≥ 1/100 to < 1/10
Uncommon:	≥ 1/1,000 to < 1/100
Rare:	≥ 1/10,000 to < 1/1,000
Very rare:	< 1/10,000
Not known:	Frequency cannot be estimated from the available data

Immune System Disorders
Rare: Hypersensitivity reactions
Not known: Anaphylactic reactions including anaphylactic shock, angioedema and pruritus

Skin and subcutaneous tissue disorders
Rare: Rash, urticaria
Not known: Severe cutaneous adverse reactions (including erythema multiforme, Stevens-Johnson syndrome (SJS)/toxic epidermal necrolysis (TEN) and acute generalized exanthematous pustulosis (AGEP)).

Gastrointestinal disorders
Common: Nausea
Uncommon: Vomiting, diarrhoea, dyspepsia, abdominal pain
Very rare: Sialorrhoea

Respiratory, thoracic and mediastinal disorders
Not known: Dyspnoea (as a symptom of a hypersensitivity reaction)

General disorders and administration site conditions
Uncommon: Fever, mucous membrane reactions

Overdosage

No specific symptoms of overdose have been reported to date. The symptoms observed in cases of accidental overdose or medication error are consistent with the known side effects at recommended dose and may require symptomatic treatment.

Availability

10's & 50's
Sustained-release capsules 75 mg
Store in a safe place out of reach of children!
Store below 30° C.
Please refer to packaging for information on shelf-life
Manufactured by
Sanofi Winthrop Industrie
196 Rue du Maréchal Juin
45200 Amilly,
France
Product License Holder:
In Malaysia:
DKSH Malaysia Sdn Bhd,
B-11-01, The Ascent, Paradigm
No.1, Jalan SS7/26A, Kelana Jaya
47301 Petaling Jaya, Selangor
Malaysia
Date of revision: Sep 2025 (CCDS V1 2019 (based on German SmPC on March 2020)

Store in a safe place out of the reach of children!

Opella.